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AI-driven corporate governance: a regulatory perspective

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ABSTRACT

The use of AI-driven or AI-assisted systems in corporate compliance is a novel topic in the business and management literature, only marginally considered by legal academia. While some scholarly pieces have explored the use of artificial intelligence (AI) systems to support compliance in the financial sector (eg, for anti-money laundering purposes), the legal implications of deploying AI systems to fulfil corporate compliance tasks has not yet received the attention it deserves. This article approaches the use of AI in corporate compliance and governance through the eyes of a regulator. Its aim is to present how the use of AI compliance systems within corporations can be leveraged by the entities tasked with ensuring the good governance of firms. Concretely, it explores a not-too-distant future, where some of the regulation of corporations can be driven by a mandate to systematically deploy Automated Compliance Management Systems (ACMS) within corporations as a way to enforce and monitor good governance. The article suggests that several of the limitations that regulators are currently facing to ensure the prevention of fraud, corruption or other forms of corporate non-compliance, could be addressed if regulators offered firms the possibility of being continuously monitored by an automated AI-driven system which systematically tracks the corporation's data footprint. We then explore what is required for such a regulatory approach to be feasible and effective. In our view, regulators must first establish the standards of reliability for those systems (ie, the legal meaning of 'red flags' issued by the system of potential violations) and, second, actively supervise how companies react to the outputs of those systems (ie, what companies do when a red flag is raised by the system). In the final part of the article, we offer a more concrete example to the reader. We suggest it could be possible to test such an approach in Australia through regulatory sandboxes for the mining sector, concretely, to monitor compliance with antibribery obligations overseas.

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Introduction

Monitoring compliance with regulatory obligations among corporations has two notable problems. First, meaningful access to corporate data by regulators is limited.¹ Corporations are generally unwilling to share access to their data footprint with regulators in the absence of legal compulsion. Yet, even if all corporate data were available to regulators, it would be impossible for any agency in a liberal democracy to adequately analyse it. The quantum of raw data produced by corporations – think of emails, transaction-related data, operational documents, inventories, accounts, human resources data, etc – generates an ‘infoglut’² that exceeds the capacity of many public agencies to manage.³

Second, when regulators decide to place that responsibility on the corporations themselves, as they are best placed to make sense of all that data and identify any risks of violation of corporate governance rules, their effectiveness can be severely reduced. Consider, for example, compliance obligations based on some strict procedures – such as *Know Your Customer*-type obligations (KYC). Regulators can readily monitor whether these KYC procedures have been adopted but may struggle to assess the level of internal enforcement, and ultimately whether the corporation has achieved the purpose of those rules (ie, to prevent money laundering). Regulators are then forced to adopt a formal approach to KYC compliance: banks can request and store identity documents and tax records of clients and meet all the procedural requirements, denying service to those who cannot meet those standards. Arguably, this approach to regulatory responsibility means that formalism (characterised as ‘regulatory ritualism’) becomes the norm, rather than effective risk-based and evidence-based approaches.⁴

This article addresses the following question: can a technology-based approach be a solution for these problems? Specifically, we explore whether automated compliance management systems (ACMS), based on artificial intelligence (AI), could be used to address these challenges. The objective of the article is to better understand the regulatory options offered by the emergence and commercial availability of AI-driven compliance management systems (CMS) – which did not exist when the current regulatory system was developed. Concretely, the article focuses on their potential contribution to meet first-order regulatory objectives such as the prevention of corporate fraud or corruption, and the mechanisms available to regulators to best exploit their potential. That is, we are less concerned about the practical implementation of the systems – an issue that would exceed the scope of this article – and explore instead what ACMS can offer in terms of both potential and pitfalls as a new device in the regulatory toolkit. Furthermore, we only tangentially address the potential risks of AI deployment – and the consequent limitations for its use in ACMS in corporate governance – as these are issues already explored at length in the legal and technical AI literature.⁵ Nonetheless, we do discuss

¹Generally, see the discussion in Cohen (2016), pp 386–387.

²Cohen (2016), p 369; more generally, see Andrejevic (2013).

³Although estimates vary, it is often assumed that corporate data storage increases by 40 per cent each year, but ‘[i]n some sectors – most notably healthcare, pharmaceuticals, energy, telecommunications, and transportation – managers report datacenter growth in excess of 100 per cent per annum’: Tallon (2013), p 33.

⁴Gill and Taylor (2004). Regulatory ritualism has been identified in a range of regulatory domains, including aged care regulation. Empirical research has revealed that compliance audits against relevant national aged care standards were paper-based (focused on medical charts and records) with no field observation of services delivered: Braithwaite, Makkai and Braithwaite (2007).

⁵Citron and Pasquale (2014) offer an early analysis of the issues raised by automation.

below how issues such as the black box effect or possible training biases interact with the operational dimension of deploying ACMS as regulatory tools. In this context, we note the many current regulatory initiatives in the AI space that try to address these and other issues, most notably the European Union's *AI Act 2024* which extends its scope to the regulation of AI deployment in the private sector.⁶

The starting point is that these automated systems, regardless of their limitations and intrinsic risks, offer regulators the possibility of (i) having *meaningful access* to corporate data; (ii) being capable of *processing* large amounts of data; and (iii) developing *effective regulatory interventions* based on these data. Yet, before turning to ACMS, it is first necessary to understand the problems highlighted above in further detail.

Precondition for effective regulation: facilitating meaningful access to corporate data

Regulators can only easily access a small part of the data held by corporations potentially relevant to their functions. The reasons for such a limited view are diverse; one significant limit is likely linked to issues of time or costs – for both the regulator and the regulated entity. Another reason corporations may push back (often legitimately) against expansive regulatory demands to share systematically more and more corporate data relates to proprietary or ‘commercial-in-confidence’ considerations.⁷ While more data may enable regulators to better perform their functions (on the basis that they would have a better understanding of the operations of regulated entity), these functions will only be possible if the regulatory agencies are capable of processing and analysing these large volumes of received data. Given the increasing number of entities subject to an ever-expanding range of regulatory supervision, the quantum of data that regulators would need to perform their functions far exceeds their current capabilities. This ‘infoglut’ problem is not likely to get any easier in the foreseeable future.

The continuous monitoring of all corporate activities by a regulator – even restricting this to surveillance of its digital footprint or ‘dataveillance’⁸ – cannot be considered a practical regulatory option unless the risks generated by the violation of the applicable rules are likely to be extreme. Consider, for example, the strict monitoring of the security rules applicable to corporations involved in developing cyberweapons. In that case a continuous supervision and testing of compliance by such a corporation with cybersecurity rules could be justified to the point of even embedding officials into the operations. Beyond these extreme cases, the current alternatives offered to regulators for corporate monitoring can be categorised into five types:

- (1) *Mandatory Reporting*. Regulators can mandate the corporation to report regularly, and then focus on those reports instead of the underlying data. Australia currently relies on this ‘mediated approach’ in relation to the *Modern Slavery Act*, under which corporations with a turnover of more than A\$100 million must comply with annual reporting requirements.⁹

⁶*EU AI Act 2024* (Regulation 2024/1689) lays down harmonised rules on AI providing developers and deployers with clear requirements and obligations regarding specific uses of AI.

⁷Weatherall et al. (2024).

⁸To use the term popularised by Roger Clarke, Clarke and Greenleaf (2017).

⁹*Modern Slavery Act 2018* (Cth), s 5.

- (2) *Outsource Monitoring*. Regulators can also outsource monitoring responsibilities over corporations – either in full or part – to third party auditors. This approach commonly involves auditors conducting some degree of sampling the underlying data in more detail. This is the way most jurisdictions tackle the need to ensure the reliability of financial reporting.¹⁰
- (3) *Risk-management*. Regulators can adopt risk-based approaches for the detection of regulatory breaches. This method considers the activity of the regulated entities, and the historic profiles of risks of violation of governance obligations, to focus monitoring activity on specific corporations, or certain areas of their operations. For example, the Organisation for Economic Cooperation and Development (OECD) deems that regulators should more carefully monitor the risk of bribery of foreign officials in the operations of companies from the signatory countries in certain industrial sectors (for example, the extractive industries) or states where this form of corruption is historically more prevalent.¹¹
- (4) *Responsive Regulation*. Regulators can also take a ‘responsive’ approach to regulation, prioritising and facilitation third-party reporting of potential violations from whistleblowers, civil society actors or competing firms.¹² This is, for example, the approach adopted by the Securities and Exchanges Commission (SEC) in the United States in relation to incentivising whistleblowers for securities violations,¹³ the European Union in cartel busting,¹⁴ or an increasing number of countries for anti-corruption purposes.¹⁵ Although responsive regulation is not a monitoring technique per se, it applies strategies that capitalise on the practical reality that regulated entities (specifically its senior officials and employees) are best placed to understand and transmit the meaningful data to regulators in connection to a potential violation of corporate rules.
- (5) *Self-Regulation to Meta-Regulation*. Regulators, particularly those which are under-resourced or lacking technical expertise, can adopt an entirely ‘hands-off’ approach to data-monitoring. Under this approach, regulators simply prescribe the processes which the corporate entity must follow. In this case, regulators have no access to the data at all, and regulate instead the processes that corporations must follow to ensure compliance with its obligations collecting, processing and assessing its own data. In the regulatory literature, this ‘regulation of self-regulation’ has been termed ‘meta-regulation’.¹⁶ This model imposes a statutory duty to report suspicious activity identified by those processes. For example, imposing on financial institutions an ongoing duty to report ‘suspicious’ activity is key to the federal *Proceeds of Crime Act 2002* (Cth) (POCA) regime in Australia,¹⁷ including providing material which may ground the ‘reasonable suspicion’ that triggers the making of freezing and restraining orders under the POCA, and may constitute the basis of the brief of

¹⁰Although the use of data mining techniques for this purpose has already some tradition, see Sharma and Panigrahi (2013).

¹¹OECD (2017), p 82.

¹²For a recent summary of responsive regulation and its impact on regulatory practice in Australia and globally, see Bottomley and Bronitt (2023), pp 363–375.

¹³Wiedman and Zhu (2023).

¹⁴Jiménez et al (2023).

¹⁵Bronitt and D’Amico (2018), pp 69–84.

¹⁶Bottomley and Bronitt (2023), p 357.

¹⁷*Proceeds of Crime Act 2002* (Cth).

evidence in the investigation of financial crimes and money laundering offences.¹⁸ Yet, far too often, this mode of regulation ends up being an exercise in formal check-box-ticking.¹⁹ This is problematic. On the one hand, the purpose of these procedures can be easily circumvented by bad faith actors within the corporation. On the other hand, these procedures may simply lead to compliance activity that increases operational costs (which can be simply passed onto consumers) and introduces inefficiencies, without tangible results. In summary, this is another form of regulatory ritualism.

Artificial intelligence for monitoring compliance

Given this landscape, the high expectations placed on AI to improve the efficiency and effectiveness of compliance activity are not surprising. Indeed, most of the literature on the use of AI for compliance stresses the importance of the efficiency gains derived from integrating AI into ACMS' within organisations.²⁰ This is understandable from the point of view of corporations. AI-based CMS' can produce similar outcomes at a lower cost than relying on humans for these too-often mechanistic exercises. Reliability, implementation and efficiency gains have also been the main foci of academic literature within this area.²¹

This article adopts a different perspective. It focuses instead on the use of AI systems for compliance purposes *from the regulator's point of view*. In line with the topic of the special issue, this paper discusses the design of regulatory approaches that can better exploit these automated compliance tools in order to prevent, detect and punish misconduct which takes place within, or is perpetrated by, corporations. This innovative approach, which is unexplored in academic literature, requires shifting the scope of the subject under study. Until now, scholarly attention has focused on the characteristics of these systems, and the concerns related to their implementation. This article focuses instead on what functions AI systems can perform to achieve the regulatory objectives and enhance corporate compliance.

Before engaging with the topic further, some clarification of terminology is needed. For the purpose of this article, and (mainly) to avoid interminable discussion over what can and cannot be considered an 'AI', we favour the use 'automated decision-making' (ADM) systems instead. As noted in the next section, the ADM terminology is increasingly being used in the literature in this domain. However, this does not mean that the output of these systems is fully achieved through automation. Sometimes, they can be described more accurately as Decision Support Systems (DSS) that can contribute to the work of humans to achieve the desired outcomes.²²

Understanding the context and the scope of automated compliance-management systems

Artificial intelligence in enterprise compliance management systems

Even if the concept of Enterprise Compliance Management Systems (ECMS) is commonly used in business settings, its appearance in academic literature appears to be

¹⁸English, Hickey and Bronitt (2023), Ch 3.

¹⁹For a review of self-regulation, and its regulatory limitations, see Bottomley and Bronitt (2023), pp 352–353.

²⁰Goltz and Mayo (2018); Butler and O'Brien (2019); Vecchio and Vieira (2021); Gyory et al (2023).

²¹See, for example, Cheng et al (2018), p 1; Kulkarni et al (2021); Rutagemwa and Patenaude (2022), p 291.

²²See, on this point, Bello y Villarino and Vijayarasa (2022), p 198; Bello y Villarino (2023), p 137.

very limited. According to a recent dissertation on the use of AI in compliance management there were only four scholarly pieces that adopted ‘ECMS’ terminology.²³ In that dissertation, Wall adopted her own concept for CMS in enterprise settings: CMS consist of the ‘cumulative actions of managing compliance within a framework of procedures and routines that enable the adherence to internal and external regulations and standards that emerge from both internal and external sources’.²⁴ Applying and extending Wall’s definition, ACMS refer to those computer-based systems that independently perform these functions (or assist in their performance), using the generated data in order to produce outputs relevant to assess levels of regulatory compliance. ACMS are, in that sense, tools to manage enterprise risk, as non-compliance would generate risks for the corporation.

Which compliance risks can be addressed through automation?

Enterprise risk management extends well beyond regulatory risks of non-compliance, though these are key elements of risk management in most business settings. In business circles, the Committee of Sponsoring Organisations of the Treadway Commission (COSO) has proposed a model and framework for managing risks that focusses on ‘culture, capabilities, and practices’.²⁵ This conception of risk makes automation in this broader space too ambitious. Culture is not something that can be easily changed through an automated system, and the diversity of risks for a corporation only offers limited scope for machine-learning (or other AI-related technologies), which would generally rely on the pattern-matching capacity of these technologies. Hence, it is generally assumed that the opportunities for integrating automation into business systems will reside in the narrower space of ‘pure’ compliance risks, that is, those types of risk for which there is less room for interpretation as to whether or not the regulatory condition or standards have been met. In relation to ‘pure’ compliance risk, computer-assisted systems offer a better opportunity for processing and supervision since regulatory compliance requirements are ordinarily well-understood.²⁶

Returning to the concept of compliance, the relevant ISO standard defines ‘compliance risk’ as follows: ‘likelihood of occurrence and the consequences of non-compliance with the organisation’s compliance obligations’, which are in turn defined as ‘requirements that an organisation mandatorily has to comply with as well as those that an organisation voluntarily chooses to comply with’.²⁷ This article only focuses on mandatory requirements as the minimum governance requirements that a corporation must fulfil, leaving aside voluntary compliance frameworks.²⁸

In this narrower context, corporations are already using automated systems to monitor and identify regulatory compliance risks, but this automation could be enhanced further by the use of AI. Advanced AI systems that look for patterns and deviations from patterns are a significant step forward from rules-based programming, already in

²³Wall (2022), p 21.

²⁴Wall (2022), p 18.

²⁵COSO (2017), p 3.

²⁶For example, Esayas and Mahler consider regulatory compliance ‘easily accessible’, although probably most legal practitioners will argue that this does not mean they are simple: see Esayas and Mahler (2015).

²⁷ISO (2021), ss 3.24 and 3.25.

²⁸See a discussion of the implications of this distinction for automation purposes in Esayas and Mahler (2015), p 279.

use in many companies. These rule-based programming systems, in which human programmers ‘write rules that enable the corporation to search for and find data indicative of [...] risk’²⁹ are (more or less) sophisticated alarms that can detect ‘suspect’ words in emails or payments to entities that are not an approved contractor, yet they are quite limited in what they can do and extremely easy to avoid when the search parameters are known by those seeking to avoid detection.

AI, understood in the simplest of terms as the performance by machines of tasks often linked to human intelligence, is assumed to have the potential to be a more efficient tool for supporting compliance management across an entire organisation, compared with rules-based systems. For example, a system with some degree of machine learning, actively searching for deviations from standard or approved patterns, would be much better suited to identify anomalies that can be linked to potential violations of legal obligations than a mere rules-based approach. Furthermore, the machine learning aspect could potentially serve as a deterrent for insiders who believe they can avoid static, rules-based compliance management systems.

How do these systems operate?

Having presented the context in which these systems operate and the function they serve, we turn now to how they can operate in practice. The best analogy for an ACMS is the use of an independent third party, who is granted access to all relevant electronic systems and data within a company for the purpose of searching across the myriad of data bytes for indicia (or possible events) that could be considered a compliance risk. Once the third party identifies a risk, it will notify the relevant responsible corporate official of the suspicious event, providing relevant data to support that alert. In an ideal world, the ACMS would also provide an intelligible explanation about how that determination was reached and the supporting reasons. More commonly, the ACMS may simply identify the suspicious event, leaving the organisation to determine whether or not to perform further compliance duties with a more thorough investigation.

Essentially adopting an ACMS into business operations is the equivalent of having an external compliance auditor embedded across the organisation that autonomously highlights suspicious data, transactions or matters of concern. The ACMS functions like a compliance officer or auditor who has been granted access to all data, passively monitoring and remaining silent until a threshold of suspicion triggers an alert. This idea of an embedded compliance officer has been used in relation to Deferred Prosecution Agreements (DPA) in the US and UK. Developed to deal with corporate crime and corruption, a DPA is a legal agreement between a prosecutor and the corporate defendant under which the prosecutor agrees to stay criminal proceedings for a defined period upon certain conditions being satisfied. DPA conditions typically include mandating cooperation with ongoing investigations into regulatory breaches, implementation of internal governance reforms, payment of fines and compensation to those adversely affected by corporate wrongdoing. It is not uncommon for the DPA to embed an ‘independent monitor’ within the corporation itself, although in some cases this role functions as an independent integrity advisor.³⁰

²⁹Shah and Rusch (2021).

³⁰Robinson, Urofsky and Pantel (2005). See generally King and Lord (2018).

These types of independent monitoring program are costly, typically designed and implemented by external commercial providers of legal and regulatory services.³¹ By contrast, ACMS are not only more cost-effective, but they also remain under the effective control and supervision of the corporation. From the corporate perspective, this can be advantageous, as it avoids the dispersion of responsibility among several actors, and limits the risk of third parties that are prioritising self-serving interests and failing to manage conflicts of interests.³² ACMS also has the added advantage that the data used for compliance monitoring would never leave the corporation, making corporations ultimately responsible for protecting that data, and prevent any complaint about potential leaks of that data to the competition of external actors. From a regulatory perspective, however, the use of ACMS could pose a risk for developing an efficient system of governance, as explained below when considering the potential limitations of ACMS.

The use of ACMS could also enhance timely responses. An AI-driven ACMS can effectively identify misconduct in a timely manner, which is one of the ‘selling points’ advertised by ACMS developers. For example, the CEO of Lextegrity, one of the companies providing this type of data-based compliance solutions, points out that ACMS offers real-time monitoring, unlike conventional auditing and monitoring that can only uncover issues of concern months or years after its occurrence, if at all. A recent report by the global law firm DLA Piper identified four functions that AI-driven compliance systems can assist:³³

- (1) *Finding possible risks.* AI systems can review large datasets in real-time, such as transactions, employee records, and social media, to find risks that humans may miss. This can help compliance officers focus on the most problematic areas. For example, AI can identify suspect transactions that may be related to fraud or money laundering warranting further investigation.
- (2) *Monitoring compliance programs.* AI systems can check if compliance programs are achieving their objectives. This can help to find areas where a program may need improvement and fix them before issues arise. For example, AI can check if employees are trained on corporate compliance policies and procedures, generating formal assessments that HR can manage, or randomly sending ‘problematic requests’ applying secret-shopper techniques.³⁴
- (3) *Providing insights into compliance data.* AI systems can analyse compliance data to give insights that can help compliance officers make better decisions. AI can find patterns and relationships that humans may not notice. For example, AI can find trends in compliance data that may show a need for changes in the overall compliance program at the level of the corporation.
- (4) *Automating compliance tasks.* AI systems can complete routine compliance tasks, such as reviewing documents and reports, which in turn liberate compliance officers to focus on more strategic tasks. For example, AI can review employee expense reports to identify potential fraud or errors. Indeed, university employees

³¹See a critical discussion of the role in Harrison (2017).

³²As we have seen, there is some difficulty for some expert third parties, to adequately serve different interests. For example, a third party can assist regulated entities evade or counter regulatory reforms, as it was the case of PwC in its work for the Australian Tax Office to limit tax avoidance by corporations: Belot (2023).

³³Benjet et al (2023).

³⁴Findley, Nielson and Sharman (2015).

in Australia may already be familiar with these alerts, as some universities experiment with the potential of these systems in SAP's Concur 'Verify' system.³⁵

At this point, we know that the use of ACMS offers many benefits for improving current compliance practice. The following section we examine how ACMS may be further used to re-imagine corporate governance.

A regulatory approach to corporations based on AI-driven compliance management systems

In the previous sections we discussed the roles of AI-driven compliance management systems, and the potential advantages for *regulated entities* (ie, corporations) in terms of efficiencies that could be gained from their deployment. What is less obvious is the potential for ACMS to be rolled out as another regulatory option for *regulatory agencies* to improve corporate compliance and governance more generally. In this section we first explore the conditions that make the use of ACMS feasible as a regulatory option, to then turn to the practical aspects of their operation.

Necessary requirements for an AI-driven CMS to be useful regulatory tool

In order to be a useful regulatory tool, AI-driven ACMS must satisfy the following two conditions:

- (1) the level of technology development will be sufficient for systems to be reliable tools to identify risks of non-compliance; and
- (2) the level of reliability of these systems could be determined *ex ante* through testing and standardisation.

These conditions, which we treat as assumptions for the following discussion, are intrinsically interconnected but respond to two different policy logics. Technological development assumes that ACMS will perform their functions adequately according to their (purported) technical specifications. These claims require validation through testing to ensure that ACMS perform to the desired level of reliability in a variety of real-world settings. Determining system reliability is of course a question of measurement. It can also be assumed that there will be several commercial providers of AI-driven ACMS competing for business; it is then important for the regulator to be confident that the various available systems can achieve and maintain a reasonable threshold of reliability. This assurance could be ensured through testing of each system by public authorities as part of their procurement process, or through the development of domestic standards (or international standards, as relevant) against which such systems could be certified *ex ante* by independent entities. Assuming these conditions are fulfilled, the adoption of an AI-driven ACMS would not be that different from due diligence-type defences incorporated into modern anti-bribery legislation, where corporations which implement effective

³⁵SAP Concur, 'Making an Impact with Verify', <https://www.concur.com/blog/article/making-impact-with-verify>, 25 July 2023.

compliance programs and systems of internal controls will be better able to alleviate or mitigate potential corporate liability for wrongdoing.³⁶

Due diligence-type defences, such as those outlined above, will encourage and support the adoption of ACMS, which will become key components of an effective risk-management and prevention system. In theory, where the systems work effectively corporate non-compliance should be an anomaly. Where non-compliance occurs, for example by circumventing existing controls, the corporation should not respond in the same manner as if the system was not in place. We can see this logic at play in anti-corruption legislation and implementation directives. Adopting a monitoring system is one of the three key conditions included in the latest revision of the US Department of Justice (DOJ) policy to grant a declination to prosecute in bribery cases; the other two conditions require immediate full disclosure of wrongdoing and ongoing cooperation with the investigation.³⁷ This approach has also informed the reforms currently being proposed by the Attorney-General's Department in Australia,³⁸ which, when enacted, will provide a statutory defence against the crime of failure to prevent foreign bribery.³⁹

Operating a regulatory approach to combat corporate non-compliance based on AI-driven CMS

Following from those two assumptions, we can envisage corporations moving to implement these systems on a voluntary basis, at the very least to minimise compliance costs and risks. However, in our view, the regulatory regime is not (yet) conducive to that evolution. For AI-driven ACMS to be a real option for corporations, as well as a feasible regulatory tool, a new legal regime must be established. In the final section of this article, we present an option to make this evolution more viable and likely in the context of anti-corruption legal obligations, but here we will look first into how this regime would operate in practice. For illustration purposes, we can offer an analogy with the use of monitoring systems as the preferred regulatory option in high-risk settings, specifically examining how monitoring systems have been incorporated in the regulation of nuclear plants.

In very simplified terms, the regulation of the use of high-risk technology, such as nuclear power plants, through monitoring requires:

- (1) *Implementing risk-prevention and risk-monitoring systems.* This involves the design, implementation and use of a system that is adapted to the level of risk of the regulated activity. In relation to safety at a nuclear plant, this could be adopting a system for receiving information from a series of sensors responsible for monitoring

³⁶This does not guarantee that all systems would reach a similar level of reliability. Ideally, the regulator should be able to explore the way different ACMS react to testing conditions and real-life deployment, to continuously refine those minimum requirements. To reach that point, some degree of interoperability and standardisation of data would be needed for comparability and consistency in the assessment of the reliability of different systems.

³⁷US Department of Justice - Criminal Division (2023).

³⁸Attorney-General's Department (2019).

³⁹The legislative history of these reforms has been tortuous, with repeated attempts that did not crystallise in any substantive reform. The Crimes Legislation Amendment (Combatting Foreign Bribery) Bill 2023 finally passed the Senate on 29 February 2024. It creates a new 'failure to prevent' foreign bribery offence for body corporates while providing a statutory defence to this offence if the corporation shows to have in place adequate procedures to prevent the commission of the foreign bribery offence.

anomalies. This system must be in place and working well before operations commenced, and its adoption is ordinarily linked to the licence to operate.

- (2) *Regular system testing.* The risk-monitoring systems must be tested and ‘stress tested’ regularly to confirm they are ‘fit for purpose’. This can be done by the regulator or by a designated independent entity, and it could be done in place or remotely. In the case of a nuclear plant, it normally involves testing the responses of the system to an emergency situation.
- (3) *Effective responses to alerts.* During tests and in real situations, operators of nuclear plants will be assessed based on how they respond to alarms for anomalies. In nuclear plants, alerts can be directly notified to the regulator, or the regulator can access the log of outputs of the systems (alerts issued).

Transposing this *modus operandi* to the domain of corporate governance, we can see how AI-driven ACMS could be used as a regulatory tool. The example below relates to corruption (namely bribery) but could be equally extended to other domains (fraud, money laundering, consumer protection, tax and environmental offences, etc).

- (1) *Implementation.* An AI-driven ACMS should be in place, with access to all the relevant data that a regulator could consider relevant to compliance matters. The ACMS must be assessed to be reliable as part of the corporation’s licence to operate.
- (2) *Testing.* The ACMS must be monitored and regularly tested. This should include, for example, introducing fake data potentially linked to corrupt practices to assess whether the system detects that anomaly. If detection fails, the system must be reassessed, and the sensitivity of the system recalibrated.
- (3) *Response.* The key element of this approach is to focus on how staff, management and board of the corporation reacts to any output generated by the ACMS that identifies potentially corrupt practices. Optimally, the system would produce a log accessible to the regulator, and active reporting could be envisaged, such as requesting that the corporation produces an annual report explaining how it responded and remediated these alerts.

In the above example, it is apparent that task of the regulator may be significantly simplified, while at the same time maintaining adequate levels of supervision. The relationship between the regulatory agency and the corporation would be limited, in normal conditions to first certifying that the ACMS has been properly implemented and meets the minimum requirements – this will be a precondition to granting the corporate licence to operate; and second, the regulator will be tasked to analyse how the corporation has responded to any alarms generated by the system.

Potential limitations to the feasibility of ACMS in corporate governance

Before moving to the final part of the article, we must consider potential limitations in the use of ACMS as a regulatory tool. Some of the limitations discussed below appear to be mainly technical in nature, and some others are relevant at the system level. We list them together as, in our view, they cannot be clearly separated. Where employees of corporations being monitored by an ACMS find a way to ‘game the system’ in order to

evade and conceal compliance breaches (and which was not revealed during system testing) the ACMS training dataset and possible biases may need to be reconsidered.

Difficulties assessing responses without evidentiary support: the 'black box' effect

The ACMS may not provide explicit reasons or sufficient context for decisions to flag certain activities as 'suspicious'. In that sense, the system shares similarities with the processes of 'suspicion-making' within reporting entities and law enforcement agencies in relation to anti-money laundering offences (AML), which remains largely invisible to outsiders (persons and businesses affected, courts, scholars and the wider community). To maintain public confidence in ACMS in this context, it is essential that the system is not a 'black box' for suspicion-making.

The output of system must then be supported by some degree of identification regarding the elements that were considered to trigger an alert. This can refer to the type of materials being analysed, and the surveillance and risk assessment tools and techniques employed. Ideally, systems will provide an indication about the people involved and who is best qualified to provide information helping compliance officers within companies to determine whether the threshold of reasonable suspicion has been reached in relation to any activity. This will also provide corporations with guidance on suitable responses and establish a framework for regulators to assess the appropriateness of the corporate response to the alert.

Biases in perceptions of the outputs or in training dataset

Another concern about machine learning from datasets of previously flagged 'suspect' transactions or conduct is that these can both reflect and embed biases. The issue of bias in relation to law enforcement and intelligence agencies is well-known, contributing to the selective surveillance and investigative attention directed to 'suspect populations'. Regulatory discretion is ubiquitous, exercised by regulators, enforcement agencies, prosecutors, and the courts at many points during the criminal process.⁴⁰ From a legal perspective, 'reasonable suspicion' implies an objective evidence-based standard to guide and limit investigation and enforcement action. But in practice officials construct the case against the suspect: the process 'case construction' is guided by institutional 'working rules', culture and priorities, as well as prejudices which may lead to racial and socio-economic profiling. As Bottomley and Bronitt have observed, discretion in regulatory contexts poses a real risk of discriminatory law enforcement, pointing to extensive empirical evidence of over-policing of the 'crimes of the powerless', which may be contrasted with the under-policing of 'crimes of the powerful'.⁴¹

There is considerable danger that AI-generated 'reasonable suspicion' provides a veneer of techno-neutrality or cloak of scientific reliability concealing the biases and prejudicial assumptions that have been built into the training dataset. Drawing upon

⁴⁰The idea of case construction, and how biases inform which cases are deemed suitable for investigation and which are not, have been extensively canvassed in the literature: see generally, Bronitt and Stenning (2011).

⁴¹Barak (2015), p.1.

past-actioned incidents generates the well-known problem of a system that is learning from a biased dataset, where outputs reflect previous biases.⁴² Second, and perhaps more interestingly in this context, there may be biases about the construction of risk and (ab)normality that may be embedded in the way the system triggers an alert. As Amicelle and Iafolla note, finding the right balance to tune the system may be complicated in an environment of dynamic interplay of surveillance with the construction of risk and (ab)normality:

On the one hand, risk-scoring is the frequency modulator of surveillance to decide where and who to look more intensively. On the other hand, the production of suspicion yokes to the infralegal distribution of normal and abnormal conduct. The definition of (ab)normality is the trigger component for surveillance to decide what to look for. Alert systems depend to a large extent on the combination settings of surveillance, risk and (ab)normality.⁴³

Initial biases about what is ‘normal’ and ‘abnormal’ in relation to corporate conduct in certain sectors (eg, health, mining, financial) may generate distortions in the way matters of concern are flagged by the system. Depending on the bias affecting the understanding of ‘abnormality’ in the design of the system, alerts can have very different meanings. As regulators will be called to assess the responses to those alerts, those system-level biases could be continuously reinforced.

The risk of capture of the ACMS by the regulated entity

As we noted above, one of the advantages of deploying ACMS is that systems and data will remain under the control of the regulated entities. However, this generates an obvious risk of capture of the ACMS by the monitored corporation.⁴⁴ The concept of ‘regulatory capture’, where regulated entities exercise undue and systemic influence over the regulator, is more prevalent where professionals working within regulated entities and regulatory agencies share similar backgrounds, values and interests. These ongoing working relationships assist with negotiation of settlements over corporate breaches. In relation to ACMS, the risk is of a different nature, since the corporation is responsible for commissioning and implementing the monitoring system. This allows for two types of capture: one is *overt* capture that forms part of corporate policy, and the second is *clandestine* capture that is driven by rogue actors within the corporation.

Overt capture occurs when the corporation has no intention from the outset to comply with its legal and regulatory obligations – it uses an ACMS that cannot detect suspect activity or feed relevant data to the system to enable its detection. This risk governing the regulatory use of ACMS can only be addressed by adherence to the first prerequisite noted above: namely, that the performance and effectiveness of the system, as authorised by the regulator, must be established *ex ante* as a condition of its ‘licence to operate’. There is still a risk that initial testing may be insufficient or inadequate. It is also possible that rogue companies may find ways to avoid raising flags in the ACMS once it is deployed. It is then essential that the standards for

⁴²For a general explanation of the problem, see Ntoutsis et al (2020); for possible ways to address this in a legal context, see Wachter, Mittelstadt and Russell (2020).

⁴³Amicelle and Iafolla (2018), p 858.

⁴⁴Shapiro (2012).

reliability are properly developed and tested, and that this is reinforced by external testing of the system by the regulator – in effect instituting systems surveillance or the monitoring of the monitoring system.

Regarding the possibility of rogue actors gaming the system, it is essential that corporations regularly test the system and that this forms part of their compliance responsibilities. Regular testing could serve several purposes. It counters the risk that ACMS might instil a laxity in vigilance amongst corporations, which may then not take compliance as an active and continuous duty. An obligation to undertake regular testing could avoid or mitigate against this laxity while generating a continuous learning process for the humans testing the systems.

Gaming the system may however be limited through the use of an AI-driven ACMS rather than a rules-driven CMS. Dynamic machine learning has the advantage that it allows systems to keep learning on the job. Hence, with new information included as part of their continuous learning, ACMS may be able to detect new or emerging patterns of corporate non-compliance that were not part of the original training dataset or identified in initial tests. The ‘black-box’ character of the system thus operates as a deterrent for rogue activity, since any systematic exploitation of blind spots would likely be detected as a new or emerging pattern of non-compliance.

Promoting passiveness from the regulator without adapting to sudden changes in circumstances

Another limitation is that regulators, assuming that the ACMS are fulfilling this role, may reduce interest in, and resources devoted to, active inspections or ‘whistleblowing’ within the regulated entity. Such regulatory complacency would be a mistake. Pro-active investigations, based on risk factors not necessarily covered by the systems, must be maintained and ‘whistleblowing’ programs should be enhanced.

The predicted efficiencies of utilising ACMS potentially enable regulators to be more strategic in their deployment of proactive investigation. Indeed, the use of ACMS enables regulators to aggregate data about alerts and incident responses from many corporations across one sector. This can allow regulators to analyse those alerts and assess the effectiveness of responses across a representative sample of corporations. Regulators would then be equipped to pinpoint suspected anomalies in patterns of alerts across the industry, as well as identify abnormal instances where corporations, compared to those with similar profiles, lack alerts.

Conversely, there is a risk that the ACMS may overreact to sudden, unforeseen changes in circumstances, triggering a surge in the number of alerts that necessitate investigation or regulatory response. For example, an extreme event, such as the coronavirus pandemic, could force corporations to change procedures in ways that could be identified by ACMS as potentially suspicious. In these cases, there may be ‘bottlenecks’ both for the corporations and the regulator to analyse all responses to alerts. For these extreme changes of circumstances, an adaptative system should be envisaged. The regulator should (at least) be allowed to alter the response requirements to these alerts, until systems could be dynamically trained with the new patterns.

Piloting an AI-driven ACMS for extractives: a regulatory sandbox approach

At this stage, the advantages of the regulatory approach outlined above may seem tangible, though some readers may remain sceptical about the feasibility of our proposal. To allay such scepticism, we propose here the deployment of a ‘regulatory sandbox’ approach.

Regulatory sandboxes are environments where entities operate under different rules to those that commonly apply to regulated firms, with regulators typically providing guidance aimed at developing a collaborative relationship between regulators and regulated entities.⁴⁵ The extractive industries could serve as an ideal testing ground for the use of regulatory sandboxes, particularly in the fight against foreign bribery.⁴⁶ Globally, one in five cases of foreign bribery occurs in the extractives sector.⁴⁷ In the Australian context, it has been identified as a problematic area, where current regulation may not be working adequately and there may be challenges to its enforcement.⁴⁸ Furthermore, mining corporations in Australia present several traits that make this approach particularly relevant for this sector:

First, there is a finite number of corporate players in the extractives sector, and, among them, several are multinational corporations, which creates an opportunity for substantial investment in sector specific ACMS, especially if these can be fine-tuned to meet the regulatory requirements of several jurisdictions. Second, in the extractives sector, there is an abundance of written processes that can provide relevant data to feed the system. Third, corporations engaged in the extractive industry are likely to make disproportionate use of independent subcontractors. The relationship with subcontractors can generate additional relevant data that also can be fed into the system, meeting regulatory obligations imposed on mining corporations but that are more directly linked to the behaviour of these subcontractors or partners (eg, consultancies). Fourth, due to the scale of its operations, mining corporations have an interest in building and maintaining effective working relations with domestic and foreign public officials since the licencing of their operations are largely determined by public sector authorisations. This makes bribery prevention particularly important for these firms. Fifth, these corporations can generate many official filings. Sixth, several of these corporations operate in market segments where input and output prices are relatively transparent: accordingly, these may also be fed into the ACMS as part of the data that could be used to generate alerts. For example, the bidding for licences and payments could be matched to market prices and futures contracts traded in public exchanges.

The deployment of ACMS can also yield significant benefits for mining corporations and the extractive sector at large. Corporations will enjoy a lower regulatory burden from the compliance perspective and may see the benefit in testing new approaches in an area where Australian expertise is at the highest global level. The incentive could be a reduction in corporate filings and other compliance requirements, which now extend as a minimum, according to OECD best practice, to monitoring employee gifts, travel

⁴⁵ Allen (2019), p 592; Braithwaite (2017), p 117.

⁴⁶ Extractive industries recover raw materials from the earth, process them, and turn them into products and services for use by consumers: see United Nations (2021).

⁴⁷ OECD (2016).

⁴⁸ OECD Working Group on Bribery (2016), pp 26–27.

and entertainment expenses; the reporting of charitable and political donations; and undertaking due diligence on third parties such as consultants, contractors and customers.⁴⁹ Deploying ACMS would also offer the advantage of decreasing the risk of regulatory capture, where the ‘interest alignment’ between the regulator and the regulated entity is capable of distorting enforcement. As highlighted in the literature, states may not be motivated to adequately enforce laws pertaining to the overseas activities of its corporations, especially where there is no direct harm to its population or domestic interests.⁵⁰ That said, having a record of flagged activities and alerts that must be addressed by corporations and, in turn, monitored by regulators would make it much more difficult to justify inaction or undue leniency.

Conclusion

The application of AI-driven or AI-assisted systems in corporate compliance remains largely uncharted territory in the business, management, and legal literature. While some academic work has delved into the use of AI systems in compliance within the financial sector – such as AML – a comprehensive exploration of the legal and regulatory implications of using AI systems for broader corporate compliance is still lacking. This article aims to fill this gap by examining the role of AI in corporate compliance and governance from a legal and regulatory standpoint. The article also examines how AI compliance systems can be employed by regulatory authorities responsible for ensuring corporate good governance, anticipating a future where corporate regulators require a systematic implementation of ACMS to enforce and monitor good governance practices.

Many of the current limitations regulators face in preventing fraud, corruption or other forms of corporate non-compliance could be addressed by offering firms continuous monitoring through an automated AI-driven system that systematically tracks the corporation’s data footprint. To make such a regulatory approach feasible and effective, regulators need first to establish *ex ante* standards of reliability for these systems (for example, ensuring the system’s interpretation of ‘red flags’ of ‘reasonable suspicion’ is free of bias and prejudice); and second, actively maintain surveillance and supervision over corporate responses to any system outputs (ie, the adequacy of corporate responses once a ‘red flag’ is raised). In the concluding part of the article, to allay reasonable concerns over the feasibility applying ACMS to corporate governance, we propose future regulatory reform should embrace the experimental design ideas of the ‘regulatory sandbox’. Using the extractives sector in Australia, and the problem of foreign bribery, we sketch how the regulated entities (mining corporations and industry bodies) could partner and pilot with relevant regulatory agencies in the development and implementation of AI-driven ACMS to enhance corporate compliance and accountability. Assuming our suggestion were to be developed and tested in a regulatory sandbox setting, it is conceivable that the pilot may be received differently by different corporations and other stakeholders. For some, the requirements to use embedded compliance officers and ACMS tools could be seen as a disincentive to business investment. Others may see it as a beneficial development, providing some degree of added guidance and predictability

⁴⁹OECD (2020), p 32.

⁵⁰Bello y Villarino (2022), p 352.

and, hence, promoting investment. Finally, competitors who are not offered this option, may see this approach as regulatory dumping, as it could selectively offer an extra layer of protection against enforcement and penalties. In that sense, deployment of these new tools, even at a pilot stage, should be carefully designed and implemented engaging as many relevant stakeholders as possible.

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